

Security+ 3.4 · Power Resiliency

Master CyberSec — Official CompTIA Security+ (SY0-701) Comprehensive Study Guide · **Lesson 3.4 — Power Resiliency (4:02)** · [Official YouTube Video ↗](#) · [Professor Messer Study Hub ↗](#)

Executive Lesson Summary: Power resiliency ensures continuous operation during power disruptions. Surge protectors guard against voltage spikes (typically for home/office). UPS (Uninterruptible Power Supply) provides battery power for minutes, allowing clean shutdown or generator startup. Generators provide power for days/weeks using diesel or gas. Data centers use redundant UPS and generators.

High-Yield Core Concepts Tested on Exam

- UPS: provides minutes of power—enough for shutdown or generator start
- Generator: provides days/weeks of power—UPS covers transition
- Surge protector: protects against voltage spikes
- Redundancy: multiple UPS and generators for critical systems
- Power resiliency prevents data loss and downtime

Exam Tips & Classic Scenarios

- 'temporary power during electrical outage' = UPS
- 'long-term power during extended outage' = generator
- Surge protector does not provide backup power

Typical Exam Scenario: A server continues running during a power surge and brief outage due to a UPS and surge protector.

Beyond the Video: Official CompTIA Exam Concepts, Protocols & Traps (Lesson 3.4 — Power Resiliency)

Why is this advanced coverage required for SY0-701? Video overviews provide foundational familiarity, but CompTIA tests practical engineering judgment and requires deep familiarity with the following protocols, acronyms, and operational nuances in this domain:

- **Principle of Least Privilege:** Users and computational entities must be granted only the minimum permissions strictly required to perform designated tasks.
- **Defense-in-Depth (Layered Security):** Application of multiple security controls (technical, administrative, physical) to prevent single-point perimeter compromise.

Classic CompTIA Traps & Question Framing:

- Avoid rote memorization: CompTIA exam prompts regularly adjust technical constraints requiring you to defend why one valid security control is the *MOST appropriate* or *FIRST step*.
- Scrutinize question qualifiers: 'without additional licensing costs' favors native OS tools; 'granular administrative command accounting' favors TACACS+ over RADIUS; 'non-repudiation' mandates asymmetric private key signing.

Practice Questions with In-Depth Solutions (10 Questions)

Detailed answer key with step-by-step technical rationale and comprehensive distractor breakdown provided below. Attempt answering independently first.

1. RTO (Recovery Time Objective) measures:

- A)** How much DATA you accept losing
- B)** How much TIME the service can be down
- C)** The cost of the incident
- D)** The number of affected users

2. Corporate scenario (Power Resiliency): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 3.4), an analyst evaluates the following technical requirement: 'UPS: minutes — allows time for clean shutdown / generator to start'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Implement the corresponding solution: UPS: minutes — allows time for clean shutdown / generator to start...
- D)** Allow shared access among administrators with generic credentials

3. Corporate scenario (Power Resiliency): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 3.4), an analyst evaluates the following technical requirement: 'Generator: days/weeks — the UPS covers the transition'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Implement the corresponding solution: Generator: days/weeks — the UPS covers the transition...
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators using generic credentials

4. Corporate scenario (Power Resiliency): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 3.4), an analyst evaluates the following technical requirement: '"temporary power during electrical outage'=UPS'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Implement the corresponding solution: 'temporary power during electrical outage'=UPS...
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators with generic credentials

5. Corporate scenario (Power Resiliency): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 3.4), an analyst evaluates the following technical requirement: 'server does not stop at power peak'=UPS+surge.'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Allow shared access among administrators with generic credentials
- D)** Implement the corresponding solution: 'server does not stop at power peak'=UPS+surge...

6. Corporate scenario (Power Resiliency): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 3.4), an analyst evaluates the following technical requirement: 'UPS: minutes — allows time for clean shutdown / generator to start'. What is the best recommendation or applicable control classification?

- A)** Implement the corresponding solution: UPS: minutes — allows time for clean shutdown / generator to start...
- B)** Temporarily disable audit logs to accelerate system throughput
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators using generic credentials

7. Corporate scenario (Power Resiliency): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 3.4), an analyst evaluates the following technical requirement: 'Generator: days/weeks — the UPS covers the transition'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Implement the corresponding solution: Generator: days/weeks — the UPS covers the transition...
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators with generic credentials

8. Corporate scenario (Power Resiliency): During a security review and compliance audit aligned with the CompTIA Security+ (SY0-701, Section 3.4) exam, an analyst evaluates the following technical requirement: "temporary power during electrical outage'=UPS'. What is the best recommendation or applicable control classification?

- A)** Implement the corresponding solution: 'temporary power during electrical outage'=UPS...
- B)** Temporarily disable audit logs to accelerate system throughput
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators with generic credentials

9. Corporate scenario (Power Resiliency): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 3.4), an analyst evaluates the following technical requirement: "server does not stop during power spikes'=UPS+surge.'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to speed up system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Allow shared access among administrators with generic credentials
- D)** Implement the corresponding solution: 'server does not stop during power spikes'=UPS+surge....

10. Corporate scenario (Power Resiliency): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 3.4), an analyst evaluates the following technical requirement: 'UPS: minutes — allows time for clean shutdown / generator to start'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Implement the corresponding solution: UPS: minutes — allows time for clean shutdown / generator to start...
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators using generic credentials

COMPREHENSIVE ANSWER KEY & DISTRACTOR BREAKDOWN

1) B — How much TIME the service can be down.

✓ **Why it's correct:** The correct answer is: how much TIME the service can be down. RTO (Recovery Time Objective) is a central metric in the Business Continuity Plan (BCP) and Disaster Recovery Plan (DRP), covered in the Business Continuity and Disaster Recovery domain of CompTIA Security+ SY0-701. It defines the maximum acceptable time to restore a service or process after an interruption, before the business impact becomes unacceptable. For example, if an e-commerce site sets an RTO of 2 hours, the recovery team must restore operations within 2 hours after the incident. RTO guides the choice of recovery strategies (hot site, warm site, cold site), backup frequency, and investments in redundancy. It is often confused with RPO, but while RTO measures downtime, RPO measures data loss.

✗ **Why the others are wrong:**

- *How much DATA you accept losing:* This is the definition of RPO (Recovery Point Objective), not RTO. RPO establishes the maximum amount of data that can be lost, measured in time (e.g., 15 minutes of transactions). It determines backup frequency: if the RPO is 1 hour, backups must occur at least every hour. RTO, on the other hand, focuses on service restoration time, not data loss. Confusing the two concepts leads to inadequate recovery strategies, as a low RPO does not guarantee a low RTO, and vice versa.
- *The cost of the incident:* The cost of the incident is a financial metric that quantifies direct and indirect losses resulting from an interruption (lost revenue, fines, productivity, reputational damage). Although it is an input to justify investments in continuity, it is not the definition of RTO. RTO is a time goal, not a monetary value. Using cost as the definition of RTO would divert focus from recovery planning, which needs clear temporal parameters to guide technical actions.
- *The number of affected users:* The number of affected users is an impact metric, useful for prioritization and communication, but it does not define RTO. RTO is a recovery time objective, regardless of the number of users. A service critical to few users may have a short RTO, while a service for many users may have a longer RTO, depending on business criticality. Therefore, this alternative does not correspond to the formal concept of RTO.

2) C — Implement the corresponding solution: UPS: minutes — allows time for clean shutdown / generator to start....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 3.4 (Power Resiliency): UPS: minutes — allows time for clean shutdown / generator to start.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user behavior, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators with generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who made critical changes or malicious commands.

3) B — Implement the corresponding solution: Generator: days/weeks — the UPS covers the transition....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for

lesson 3.4 (Power Resiliency): Generator: days/weeks — the UPS covers the transition. A generator provides long-term power during extended outages, while an uninterruptible power supply (UPS) provides short-term power to bridge the gap until the generator starts.

✗ Why the others are wrong:

- *Temporarily disable audit logs to accelerate system throughput*: Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence*: Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators using generic credentials*: Destroys individual accountability and non-repudiation. It makes it impossible to prove who performed critical changes or malicious commands.

4) B — Implement the corresponding solution: 'temporary power during electrical outage'=UPS....

✓ Why it is correct: According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 3.4 (Power Resiliency): 'temporary power during electrical outage'=UPS.

✗ Why the others are wrong:

- *Temporarily disable audit logs to accelerate system throughput*: Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence*: Informal terms are administrative controls, rely solely on user behavior, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators with generic credentials*: Destroys individual accountability and non-repudiation. Makes it impossible to prove who made critical changes or malicious commands.

5) D — Implement the corresponding solution: 'server does not stop at power peak'=UPS+surge....

✓ Why it is correct: According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 3.4 (Power Resiliency): 'server does not stop at power peak'=UPS+surge.

✗ Why the others are wrong:

- *Temporarily disable audit logs to accelerate system throughput*: Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence*: Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators with generic credentials*: Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

6) A — Implement the corresponding solution: UPS: minutes — allows time for clean shutdown / generator to start....

✓ Why it is correct: According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 3.4 (Power Resiliency): UPS: minutes — allows time for clean shutdown / generator to start.

An Uninterruptible Power Supply (UPS) provides short-term battery power during an outage, giving systems time to shut down gracefully or for backup generators to start. The appropriate action is to implement the corresponding solution, which includes deploying UPS units for critical systems.

✗ Why the others are wrong:

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls that rely solely on user behavior and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators using generic credentials:* Destroys individual accountability and non-repudiation. It makes it impossible to prove who performed critical changes or malicious commands.

7) B — Implement the corresponding solution: Generator: days/weeks — the UPS covers the transition....

✓ **Why it is correct:** According to the official CompTIA Security+ exam objectives (SY0-701) for lesson 3.4 (Power Resiliency): Generator: days/weeks — the UPS covers the transition. A generator provides long-term power (days to weeks) during extended outages, while an Uninterruptible Power Supply (UPS) provides short-term battery power to bridge the gap until the generator starts.

✗ Why the others are wrong:

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to incident detection and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls that rely solely on user conduct and do not block automated attacks, injections, or runtime malware.
- *Allow shared access among administrators with generic credentials:* Destroys individual accountability and non-repudiation. It makes it impossible to prove who performed critical changes or malicious commands.

8) A — Implement the corresponding solution: 'temporary power during electrical outage'=UPS....

✓ **Why it's correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 3.4 (Power Resiliency): 'temporary power during electrical outage'=UPS. A UPS provides temporary power during an electrical outage, allowing systems to shut down gracefully or continue operating until generators start.

✗ Why the others are wrong:

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detect incidents and violates compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user behavior, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators with generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who made critical changes or malicious commands.

9) D — Implement the corresponding solution: 'server does not stop during power spikes'=UPS+surge....

✓ **Why it's correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 3.4 (Power Resiliency): 'server does not stop during power spikes'=UPS+surge..

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to speed up system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators with generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who made critical changes or malicious commands.

10) B — Implement the corresponding solution: UPS: minutes — allows time for clean shutdown / generator to start....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 3.4 (Power Resiliency): UPS: minutes — allows time for clean shutdown / generator to start.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user behavior, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators using generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.